

Zixuan Wei

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EDUCATION

Southern University of Science and Technology (SUSTech)

Sep 2024 - Jun 2028

B.Eng. in Automation School of Automation and Intelligent Manufacturing

- Major GPA: 3.88/4.00 | Major Rank: 6/78
- Undergraduate researcher at AMAS Lab under the supervision of Prof. Zhiyun Lin, focusing on robot perception, state estimation, and autonomous systems. Research experience spans 3D perception, 6-DoF pose estimation, learning-based visual state estimation, mobile robot localization and navigation, ROS-based system integration, and real-robot deployment.
- Manuscript: Second-author paper submitted to IEEE Transactions on Control Systems Technology (TCST); first-round peer review completed.
- Research Interests: Robot state estimation, temporal representation learning, 3D perception, robot learning, and embodied intelligence.
- Relevant Coursework: Modern Control Theory, Automatic Control Theory, Probability and Mathematical Statistics, Signals and Systems, Artificial Intelligence and Machine Learning, Convex Optimization and Its Applications in Machine Learning, System Modeling and Simulation, Intelligent Sensing and Signal Processing, Microcomputers and Embedded Systems.

RESEARCH EXPERIENCE

FAST-LIO-Based Mobile Robot Localization and Navigation

Sep 2025 - Feb 2026

Developed and deployed a complete mobile-robot localization, mapping, and autonomous navigation system based on ROS Noetic, Livox MID-360, and FAST-LIO for indoor autonomous navigation.

- Built a real-robot localization and navigation system integrating Livox MID-360 LiDAR and FAST-LIO, supporting 3D point-cloud mapping and real-time odometry estimation.
- Developed the interface between FAST-LIO and the 2D navigation stack by converting 3D point-cloud maps into 2D occupancy-grid maps and implementing PointCloud2 → LaserScan conversion and 3D-to-2D coordinate projection.
- Integrated global/local path planning, cost maps, real-time obstacle avoidance, and TF-based coordinate-frame management using move_base, unifying map, odom, and robot body frames.
- Connected the navigation stack with the low-level chassis controller to establish an end-to-end LiDAR sensing → localization → mapping → planning → obstacle avoidance → motion control pipeline on a physical mobile robot.

6-DoF Target Pose Estimation with a 3D Point-Cloud Transformer

Feb 2026 - Jun 2026

Developed a 3D point-cloud-based target perception and 6-DoF relative pose estimation system for a mobile robot based on PIRATR, covering synthetic data generation, model optimization, evaluation, and real-robot deployment.

- Built a 3D target perception system using Livox MID-360 + TRACER MINI, enabling target instance recognition and 6-DoF relative pose estimation, together with coordinate transformation from the world frame to the robot frame.
- Constructed a controllable 3D simulation environment using Blender + Docker, including virtual LiDAR scanning, randomized scene generation, and automated annotation, generating approximately 10,000 synthetic point-cloud samples.
- Addressed the difficulty of learning accurate target orientation caused by rotational symmetry by redesigning the target-relative geometric representation and modifying PIRATR's original rotation-symmetry loss constraint, improving the model's ability to distinguish target orientations.

- Improved the learning of geometric orientation features through target-region point-cloud densification, point-cloud resampling, random-line augmentation, high-frequency perturbation, and reweighting of pose-related loss terms; further optimized the training process using pretrained PIRATR weights and multi-stage parameter tuning.
- Performed data parsing, class encoding, label-format adaptation, checkpoint comparison, and overfitting analysis to resolve mismatches between the original PIRATR data pipeline and the target-specific single-object scenario; achieved a median yaw error of 6.84° and a mean yaw error of 11.75° on the test set.
- Deployed the trained model on a GPU server and implemented a real-robot online inference pipeline using ROS + Docker + bidirectional network communication, enabling real-time MID-360 point-cloud upload, inference, and target-pose feedback.

Gated-Kalman-UNet for Visual State Estimation: Mechanistic Analysis

Jul 2026 - Present

Investigated recursive state prediction and observation-update mechanisms in latent space for camera motion estimation from sequential visual inputs. Focused on how historical states contribute to current state estimation, and quantified the effects of different historical-state perturbations and update mechanisms on estimation accuracy by comparing predicted 6-DoF poses with ground-truth poses.

- Conducted visual odometry experiments on the KITTI Odometry Dataset, using sequences of six consecutive RGB frames with five recursive state updates to estimate 6-DoF relative poses consisting of 3D rotation and 3D translation.
- Systematically analyzed the recursive state-estimation pipeline in Kalman-UNet / Gated-Kalman-UNet: previous posterior state $\rightarrow$ current predicted prior $\rightarrow$ observation-update residual $\rightarrow$ current posterior state $\rightarrow$ pose output, and compared the mechanisms of direct residual updating and adaptive gated updating.
- Designed historical-state intervention and counterfactual control experiments in which historical states were replaced while keeping the current observation and model parameters fixed, in order to quantify how changes in historical information affect the current 6-DoF pose prediction.
- Performed intermediate-variable auditing and component ablation experiments to trace how perturbations in historical states progressively affect the current predicted prior, observation-update residual, and final posterior state, and analyzed the actual roles of adaptive gating and cross-attention in the propagation of historical information.
- Conducted both static source-code auditing and runtime tensor tracing of the cross-attention modules, examining the origins and actual tensor shapes of Q/K/V, the multi-head attention structure, token dimensions, and attention-matrix computation, in order to verify the correspondence between the implemented model and its theoretical mechanism.

Technical Skills

- Robotics: ROS Noetic, FAST-LIO, move_base, TF, Livox MID-360, TRACER MINI
- Deep Learning & Perception: Transformer-based 3D perception, attention mechanisms, 6-DoF pose estimation, visual odometry, learning-based state estimation
- Simulation & Deployment: Blender, Docker, synthetic data generation, GPU-based inference, real-robot deployment
- Programming : C/C++, Python

Honors & Awards

Outstanding Student Scholarship, Southern University of Science and Technology, 2025–2026

ADDITIONAL INFORMATION

Visiting Research Availability: Seeking a one-semester Visiting Research Student (Non-Exchange) placement at NUS for Spring 2027 through SUSTech's university-level visiting research program.